

Electrical resistivity surveys to delineate groundwater potential aquifers in Peddavanka watershed, Anantapur District, Andhra Pradesh, India

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Abstract Vertical electrical resistivity soundings were conducted in order to delineate groundwater potential aquifers in Peddavanka watershed, which is a catchment of about 398 km² in Anantapur District, Andhra Pradesh, India. The main lithologic units in the watershed are quartzite, limestone, shale, and alluvium. Ninety-nine vertical electrical soundings were conducted using the Schlumberger configuration, covering the entire watershed. The data were interpreted with the help of master curves and auxiliary point charts. Interpretations of VES were used to generate a top layer apparent resistivity contour map and longitudinal conductance map. Isoresistivity contour maps were prepared and interpreted in terms of resistivity and thickness of various sub-surface layers using computer software (SURFER), and isocontour diagrams depicting the depth to bedrock were prepared. Resistivity results were correlated with the existing lithology. Based on the depth to bedrock, the thickness of the saturated layer and the resistivity of the second layer, a groundwater potential map has been prepared, in which good, moderate, and poor zones are classified. The study reveals that the weathered and fractured portions in shale and limestone that occur in the southernmost and central portions of the watershed area constitute the productive water-bearing zones categorized as good groundwater potential aquifers.

Keywords Electrical resistivity · Longitudinal conductance · Vertical electrical soundings · Groundwater potential aquifers · Andhra Pradesh · India

Introduction

Groundwater exploration is gaining more and more importance in India owing to the ever increasing demand for water supplies, especially in areas with inadequate surface water supplies. The watershed area under study is a chronic drought-prone area in Andhra Pradesh. Due to differential weathering and fracturing, the aquifers in the hard rock terrain show wide vertical and horizontal variations in their hydrologic characteristics. Many dug wells that were sunk in the study area without an initial proper investigation failed, and so were abandoned. A systematic and scientific approach to the problem is therefore essential for the study area in order to overcome these problems.

Hard rock areas are known to yield limited quantities of groundwater due to their mode of origin and the absence of intergranular porosity. The extreme drought conditions, depletion of surface water, and large fluctuations in the water table prevailing in hard rock terrains all emphasize the need to locate potential groundwater aquifers in the study area. The Peddavanka watershed (Fig. 1) is formed by four ephemeral streams with an area of 398 km², and lies between latitudes 14°55'24"–15°12'50" N and longitudes 77°51'20"–78°03'26" E. The mean, maximum, and minimum rainfall in the area is 515 mm, 1285 mm, and 217 mm, respectively for a 94 year period (1901–1994). While horizontal geoelectrical sections at different depths throw light on the aerial distribution of different groundwater potential aquifers in general (Yadav and Lal 1989), integrated studies involving geological, hydrological, meteorological, water balance and geophysical surveys are useful for identifying potential groundwater aquifers in the study area.

Geology

Most of the watershed area is occupied by the Tadipatri formation of the Lower Cuddapah Super group, which is mainly represented by shale with interbedded layers of

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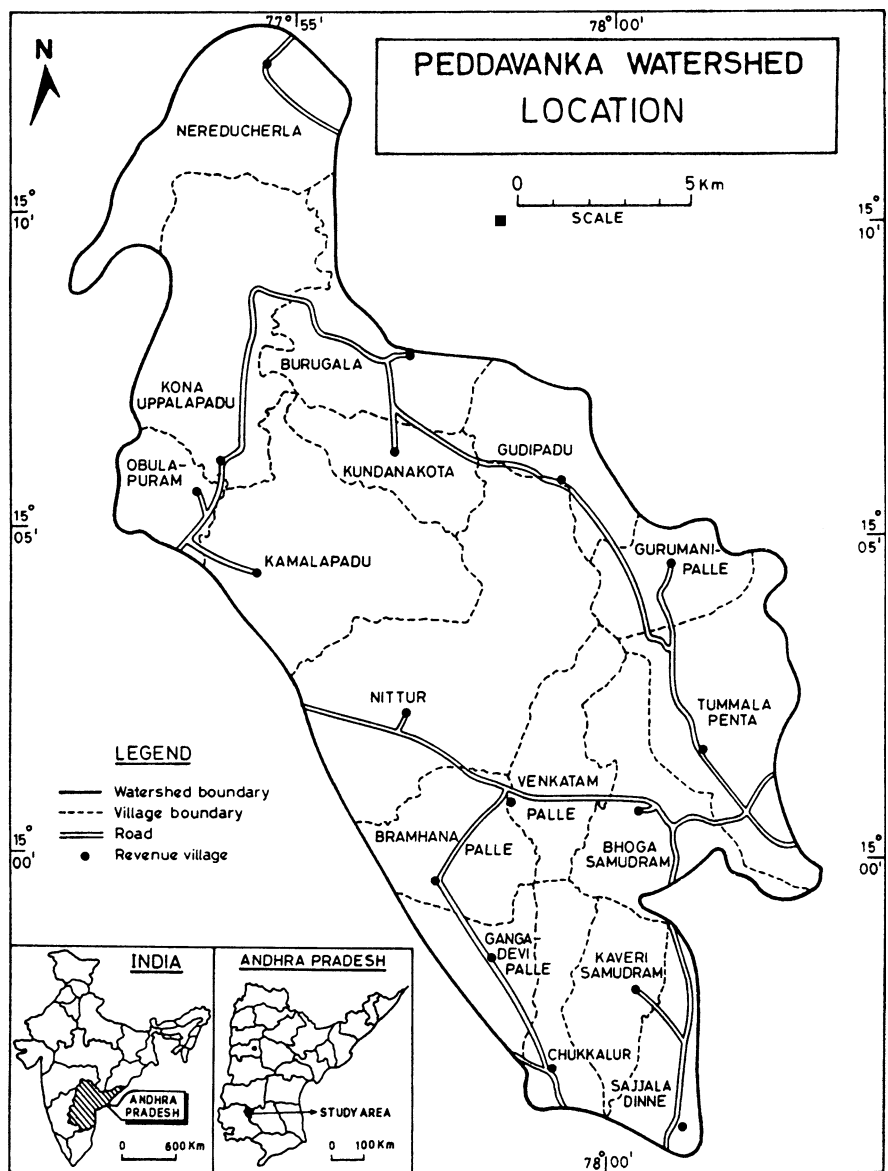


Fig. 1
Map showing the location of Peddavanka Watershed, Andhra Pradesh, India

flaggy limestone with quartzite as an outlier. The shale trends NNE-SSW with an $\sim 10^\circ$ dip towards the east. Shale often contains layers of basic volcanic rocks, mainly sills of trap and trap conglomerate with volcanic pellets. The Tadipatri formation is overlain by the rocks of the Kurnool group, which include Banaganapalle quartzite, Narji limestone, and Auk shale. A distinct angular unconformity separates the

Cuddapah formation from the Kurnool rocks, the latter occurring as horizontal beds over the tilted Cuddapah rocks. Alluvium of recent age comprises sand, silt, and clay, which occur all along the course of the stream and at the confluence of the stream with the Pennar River in the southern portion of the watershed (Fig. 2). The geological succession in the watershed is as follows:

Kurnool Group	Paniam Quartzite	(10–35 m)
	Local disconformity	
	Auk Shale	(10–30 m)
	Narji Limestone (massive limestone & flaggy limestone)	(100–200 m)
	Banaganapalle Quartzite (and Conglomerate)	(10–57 m)
Cuddapah Super Group	Unconformity	
	Chitravati Group	
	Tadipatri formation (4600 m)	
	Shale and basic intrusives: sills	

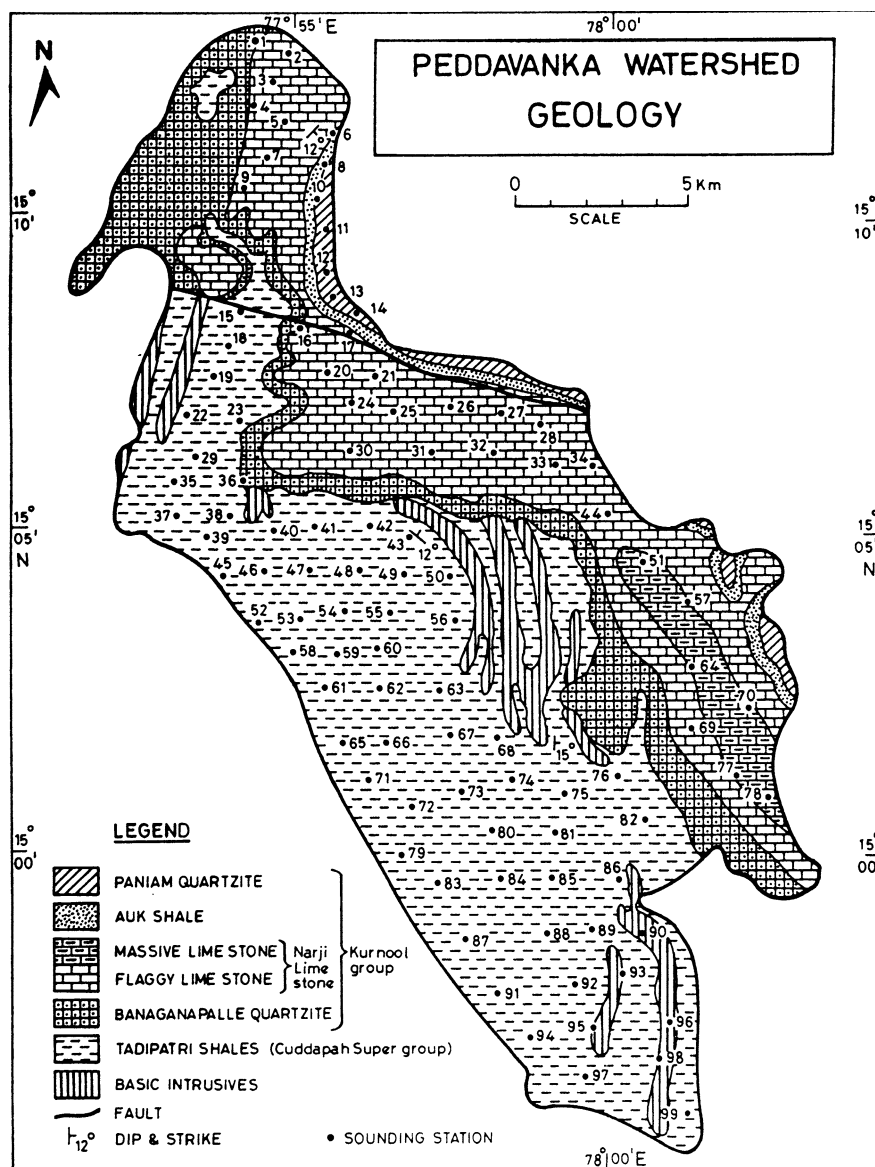


Fig. 2
Geological map of the study area showing vertical electrical sounding stations

Electrical resistivity surveys

The electrical resistivity method, of all the surface geophysical methods, has been applied most widely in groundwater exploration studies (Todd 1995). The literature available on electrical resistivity prospecting is described in various books (Keller and Frischknecht 1966; Bhattacharya and Patra 1968; Zhdanov and Keller 1994; Appa Rao 1997). The electrical resistivity method can be best employed to estimate the thickness of overburden and also the thickness of weathered/fractured zones with reasonable accuracy. Though both Wenner and Schlumberger electrode configuration methods are popularly employed, the Schlumberger electrode configuration method is more suited to the study area, ensuring better results. The method has practical, operational, and interpretational advantages over other methods such as the Wenner method of electrode arrangement (Zohdy and others 1974; Bhimasankaram and Gaur 1977). An indigenous portable

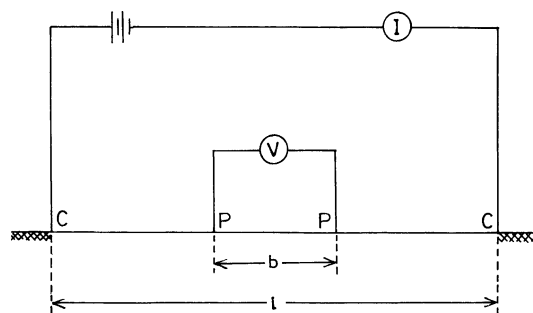


Fig. 3
Schlumberger electrode arrangement

D.C. earth resistivity meter was used for electrical probing. The Schlumberger electrode configuration was employed, with a maximum current electrode (AB/2) separation of 90 m. In the Schlumberger method, the four electrodes are arranged in such a way that the distance between the two inner potential electrodes (b) is kept small compared to

Table 1
Results of interpretation of vertical electrical soundings

VES number	Longi- tude	Latitude	Type of curve	ρ_1	h_1	ρ_2	h_2	ρ_3	h_3	ρ_4	Depth to basement “H”
1	77°54'12"	15°12'42"	K	23	1.7	897	22.1	228	6.7	–	30.5
2	77°54'50"	15°12'32"	K	19	1.4	741	28.0	318	8.5	–	37.9
3	77°54'33"	15°12'02"	K	30	1.6	1170	17.6	323	8.9	–	28.1
4	77°54'20"	15°11'38"	K	28	1.5	1092	18.0	294	4.8	–	24.3
5	77°54'46'	15°11'28"	A	11	1.6	99	25.6	1520	–	–	26.6
6	77°54'38"	15°11'20"	KH	70	1.4	130	0.6	56	7.0	2340	9.0
7	77°54'32"	15°10'55"	A	11	1.4	99	16.8	7227	–	–	18.2
8	77°55'32"	15°10'50"	H	630	1.5	510	5.0	–	–	–	6.5
9	77°54'13"	15°10'30"	A	8	1.3	160	19.5	–	–	–	20.8
10	77°55'22"	15°10'16"	KH	55	1.4	128	1.8	37	3.8	4950	7.0
11	77°55'32"	15°09'45"	A	35	1.6	700	9.0	–	–	–	10.6
12	77°55'32"	15°09'08"	A	34	1.9	680	9.5	–	–	–	11.4
13	77°55'38"	15°08'38"	A	32	1.6	1280	16.0	–	–	–	17.6
14	77°56'03'	15°08'17"	A	31	1.5	1240	15.0	–	–	–	16.5
15	77°54'10"	15°08'18"	H	25	1.5	11	9.5	247	–	–	11.0
16	77°55'05"	15°08'12"	A	11	2.1	220	10.5	–	–	–	12.6
17	77°55'52"	15°08'03"	A	10	1.4	250	11.2	–	–	–	12.6
18	77°53'57"	15°07'52"	KH	31	1.7	45	2.6	14	11.3	147	15.6
19	77°53'47"	15°07'28"	HK	32	1.9	11	11.0	117	46.2	232	59.1
20	77°55'33"	15°07'32"	A	8	2.1	320	31.5	–	–	–	33.6
21	77°56'14"	15°07'28"	A	8	2.0	320	30.0	–	–	–	32.0
22	77°53'18"	15°06'48"	AK	2	1.7	40	8.5	153	19.0	323	29.2
23	77°54'10'	15°06'41"	H	23	2.1	10	13.2	228	–	–	15.3
24	77°55'50"	15°07'00"	A	120	3.1	840	10.8	1165	–	–	13.9
25	77°56'40"	15°06'54"	A	105	2.7	588	14.8	1230	–	–	17.5
26	77°57'38"	15°06'52"	KH	98	1.2	147	1.6	61	10.5	2535	13.3
27	77°58'20"	15°06'46"	KH	90	1.5	135	1.2	59	8.4	2730	11.1
28	77°58'58"	15°06'44"	A	48	1.5	72	12.0	265	–	–	13.5
29	77°53'31"	15°06'12"	A	2	1.6	34	8.0	–	–	–	9.6
30	77°55'47"	15°06'26"	A	95	2.3	950	23.0	190	–	–	25.3
31	77°57'07"	15°06'18"	AA	11	2.1	33	4.2	131	54.4	–	60.7
32	77°58'06"	15°06'17'	A	160	1.9	296	17.1	1624	–	–	19.0
33	77°58'53"	15°06'11"	AA	170	2.0	255	5.0	960	26.0	1750	33.0
34	77°59'36"	15°06'03"	AA	170	2.1	255	4.6	690	24.7	825	31.4
35	77°53'12"	15°05'50"	A	32	2.6	80	26.0	–	–	–	28.6
36	77°54'07"	15°05'50"	A	2	2.2	48	6.6	–	–	–	8.8
37	77°53'10"	15°05'17"	KH	25	2.0	100	4.0	52	6.7	558	12.7
38	77°53'58"	15°05'19"	KH	34	2.6	136	1.8	35	5.8	5445	10.2
39	77°53'36"	15°04'58"	HA	33	1.9	270	6.8	30	20.0	112	28.7
40	77°54'36"	15°05'04"	KH	15	1.7	18	2.0	12	10.6	142	14.3
41	77°55'18"	15°05'03"	KH	17	1.6	31	1.3	15	11.5	141	14.4
42	77°55'57"	15°05'02"	HA	10	1.8	5	2.2	21	9.1	–	13.1
43	77°56'40"	15°04'55"	HA	9	1.8	3	1.4	159	8.4	–	11.0
44	78°00'04"	15°05'20"	A	7	2.0	260	20.0	–	–	–	22.0
45	77°53'48"	15°04'27"	QH	32	1.8	30	4.8	12	27.7	546	34.3
46	77°54'26"	15°04'27"	QH	80	1.3	20	8.7	13	18.0	126	28.0
47	77°55'09"	15°04'23"	KH	16	1.5	24	0.7	10	16.6	195	18.8
48	77°55'54"	15°04'20"	A	2	2.0	40	8.0	–	–	–	10.0
49	77°56'40"	15°04'17"	A	4	2.0	72	6.0	–	–	–	8.0
50	77°57'27"	15°04'17"	H	34	1.7	11	11.0	585	–	–	18.7
51	78°00'35"	15°04'35"	A	13	2.2	520	11.0	–	–	–	13.2
52	77°54'13"	15°03'52"	QH	25	1.4	20	4.0	10	9.1	468	14.5
53	77°54'53'	15°03'52"	A	24	1.8	29	5.6	84	9.4	468	16.8
54	77°55'40"	15°03'52"	KH	2	1.5	12	26.0	11	15.9	–	20.0
55	77°56'33"	15°03'47"	A	3	1.8	8	9.0	–	–	–	10.8
56	77°57'32"	15°03'40"	A	3	3.2	15	9.6	60	–	–	12.8
57	78°01'15"	15°03'58"	A	6	1.9	232	19.0	–	–	–	20.9
58	77°54'56"	15°03'17"	KH	21	1.7	25	6.8	17	39.1	361	47.6
59	77°55'40"	15°03'11"	A	2	2.0	23	6.0	–	–	–	8.0
60	77°56'20"	15°03'07"	A	5	2.1	54	21.0	–	–	–	23.1
61	77°55'28"	15°02'42"	A	3	3.0	28	30.0	–	–	–	33.0
62	77°56'25"	15°02'38"	A	3	2.9	27	29.0	–	–	–	31.9
63	77°57'17"	15°02'37"	A	5	2.8	50	28.0	–	–	–	30.8
64	78°01'20"	15°02'52"	A	10	1.7	40	8.5	–	–	–	10.2

Table 1
(Contd.)

VES number	Longitude	Latitude	Type of curve	ρ_1	h_1	ρ_2	h_2	ρ_3	h_3	ρ_4	Depth to basement "H"
65	77°55'42"	15°01'57"	A	3	2.2	14	11.0	156	–	–	13.2
66	77°56'29"	15°01'55"	A	4	2.3	18	11.5	70	–	–	13.8
67	77°57'23"	15°01'57"	A	5	2.5	50	12.5	–	–	–	15.0
68	77°58'15"	15°01'47"	KH	23	2.4	28	3.8	22	33.6	324	39.8
69	78°01'18"	15°02'05"	A	11	1.7	420	8.5	–	–	–	10.2
70	78°02'12"	15°02'10"	A	9	1.6	360	8.0	–	–	–	9.6
71	77°56'03"	15°01'26"	A	3	2.1	16	6.3	64	–	–	8.4
72	77°56'49"	15°01'04"	A	3	3.2	30	6.4	60	–	–	9.6
73	77°57'33"	15°01'13"	A	2	2.5	12	25.0	–	–	–	27.5
74	77°58'22"	15°01'15"	KH	3	1.5	13	1.2	5	6.4	220	9.1
75	77°59'10"	15°00'58"	H	22	1.5	9	6.3	99	–	–	7.8
76	78°00'08"	15°01'08"	A	2	2.5	21	12.5	42	–	–	15.0
77	78°02'03"	15°01'20"	A	11	1.8	440	9.0	–	–	–	10.8
78	78°02'38"	15°01'00"	A	11	1.7	424	8.5	–	–	–	10.2
79	77°56'40"	15°00'23"	A	2	1.8	32	9.0	–	–	–	10.8
80	77°58'02"	15°00'34"	A	4	2.1	11	10.5	–	–	–	12.6
81	77°59'05"	15°00'23"	A	2	2.4	18	7.2	–	–	–	9.6
82	78°00'28"	15°00'24"	A	2	1.9	34	28.5	–	–	–	30.4
83	77°57'02"	14°59'47"	A	4	2.7	22	8.1	288	–	–	10.8
84	77°58'00"	14°59'50"	A	4	2.5	40	12.5	–	–	–	15.0
85	77°58'52"	14°59'49"	A	3	2.4	14	24.0	–	–	–	26.4
86	78°00'03"	14°59'42"	A	3	2.3	28	23.0	–	–	–	25.3
87	77°57'33"	14°58'48"	A	2	1.4	12	27.0	–	–	–	28.8
88	77°58'42"	14°58'42"	A	5	2.8	22	14.0	–	–	–	16.8
89	77°59'28"	14°58'54"	A	12	2.4	30	12.5	–	–	–	15.0
90	78°00'32"	14°58'46"	A	11	2.6	55	26.0	–	–	–	28.6
91	77°58'03"	14°58'04"	A	3	2.2	27	33.0	–	–	–	35.2
92	77°59'12"	14°58'12"	A	3	2.8	19	5.6	76	–	–	8.5
93	78°00'07"	14°58'13"	A	16	3.0	40	15.0	–	–	–	18.0
94	77°58'38"	14°57'23"	A	3	2.8	17	5.6	34	–	–	8.4
95	77°59'28"	14°57'24"	QH	40	1.7	24	4.6	16	32.5	226	38.8
96	78°00'46"	14°57'26"	A	8	2.0	80	20.0	–	–	–	22.0
97	77°59'25"	14°56'48"	A	9	2.0	90	30.0	–	–	–	32.0
98	78°00'36"	14°56'57"	A	8	2.0	80	20.0	–	–	–	22.0
99	78°01'17"	14°56'17"	A	7	1.7	70	25.5	–	–	–	27.2

the distance between the outer current electrodes (l), while making sure that all of the four electrodes are placed along a straight line. The distance between the potential electrodes should never exceed more than one fifth of that of the current electrodes at any stage during the operation of the probing. The apparent resistivity of the layer is considered to be at the centre of the electrode array. The electrode arrangement of the Schlumberger configuration is shown in Fig. 3. The current electrodes are moved in successive steps after making a desired probe for further investigations; however, the potential electrodes are not moved unless the voltage observed between the potential electrodes is too small for measurement.

The apparent resistivity for a measured resistance $R=(V/I)$ is given by:

$$\rho_a = \pi \frac{\left(\frac{L}{2}\right)^2 - \left(\frac{b}{2}\right)^2}{b} \times R$$

where L is the distance between current electrodes, b is the distance between potential electrodes, and $R(=V/I)$ is the potential difference between potential electrodes/the current flowing.

The manner in which apparent resistivity values increase or decrease with each electrode separation forms the basis

for quantitative interpretation of the electrical resistivity data. Qualitative interpretation of the sub-surface resistivity distribution can be performed by observing the shape of the field curve. In the curve matching method, a curve is drawn by plotting apparent resistivity against electrode spacing, and this is interpreted by matching the field curve with master curves of two, three and four layer cases for various ratios of absolute resistivity, as given by Orellana and Mooney (1966). Accordingly, the three-layered earth can be classified into H, K, A, and Q type curves, based on shape. The distribution of resistivities of different subsurface layers is described below:

H-type: $\rho_1 > \rho_2 < \rho_3$
 K-type: $\rho_1 < \rho_2 > \rho_3$
 A-type: $\rho_1 < \rho_2 < \rho_3$
 Q-type: $\rho_1 > \rho_2 > \rho_3$

HQ-, KA-type, and so on, represent four layer curves.

Results and discussion

Ninety-nine vertical electrical soundings, randomly distributed, were conducted in the watershed, and their

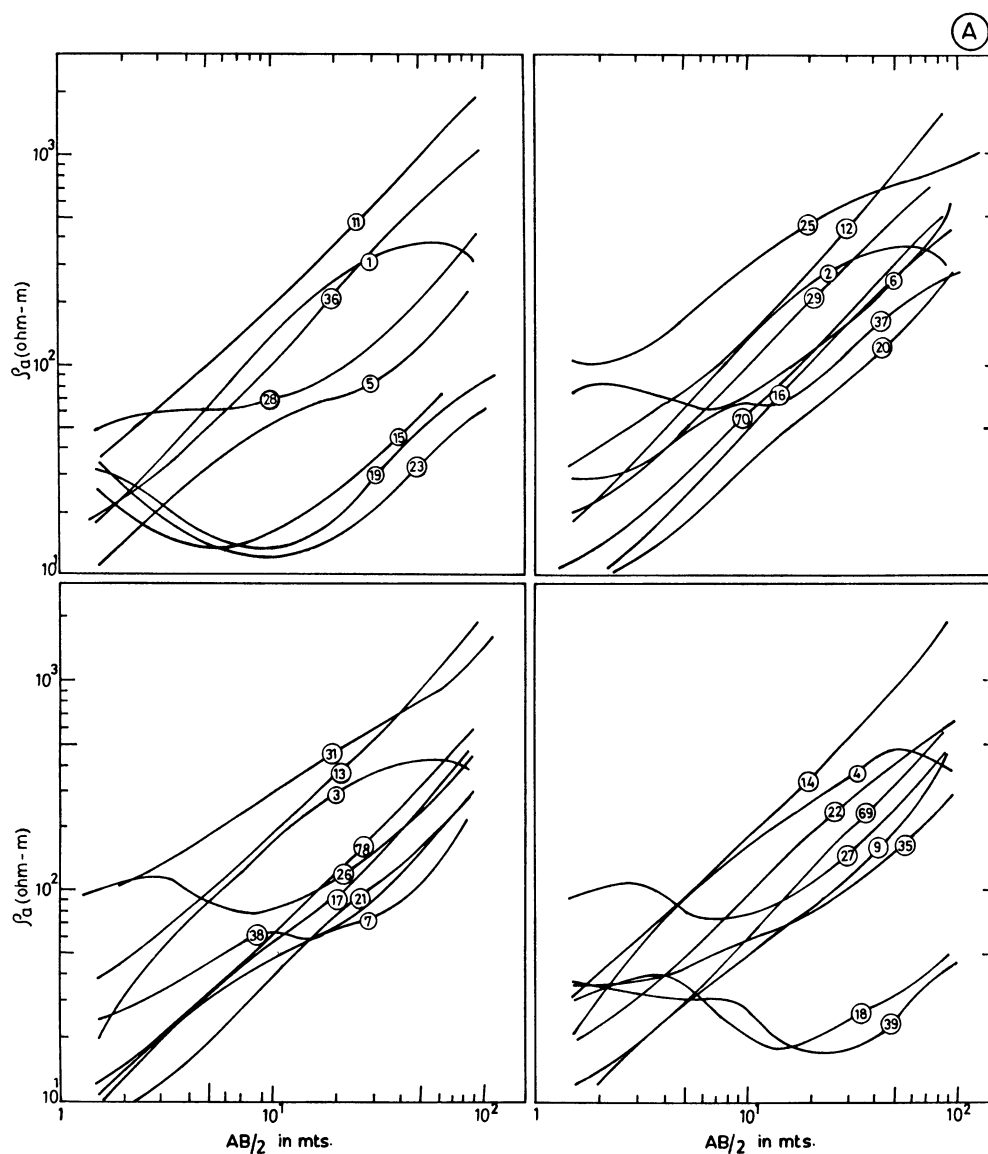


Fig. 4A–C
Vertical electrical sounding
curves

locations are given in Fig. 2. The VES data are interpreted using the curve matching technique with the help of master curves prepared by Rijkswaterstaat (1969), and the results and salient features of the subsurface parameters are given in Table 1. The sounding curves A, K, and H type suggest a few three-layer geoelectrical sections, and a number of four-layer sections of KH, HK, QH, HA, AK, and AA types (Figs. 4A,B,C). The types of curves obtained are predominantly A-type followed by KH-type, H-type, K-type, QH-type, HA-type, HK-type, and AA-type (Table 2). Apparent resistivity at different depths has been given in Table 3. Iso-resistivity contour maps at 1.5 m, 10 m, 50 m, and 90 m depths, as well as their 3-D maps, were prepared using the SURFER package, and they were interpreted in terms of resistivities and thicknesses of various subsurface layers. Based on the VES interpretation results, the top layer contour map, longitudinal conductance map, depth to bedrock map, and the groundwater potential map are prepared.

Apparent resistivity at 50 m spacing

The apparent resistivity values for 50 m spacing range from 13–949 Ωm , and the corresponding contour map is given in Fig. 5A, with a contour interval of 50 Ωm . The highest apparent resistivity value is 949 Ωm , as observed northwest of the village Burugala, between longitude 77°55'38" and latitude 15°08'38". The elevated and the low-lying areas are in almost perfect conformity with the apparent resistivity contour maps. The two peaks representing the quartzite in the northwestern portion of the basin are indicated by apparent resistivities 949 Ωm and 946 Ωm . The rest of the area is devoid of quartzite and is occupied by shale and limestone, indicated by low apparent resistivities.

Top layer apparent resistivity

The surface layer contour map is given in Fig. 6A. The resistivity value ranges from 2–630 Ωm , while its thickness ranges from 1.3–3.2 m. The variation in resistivity is

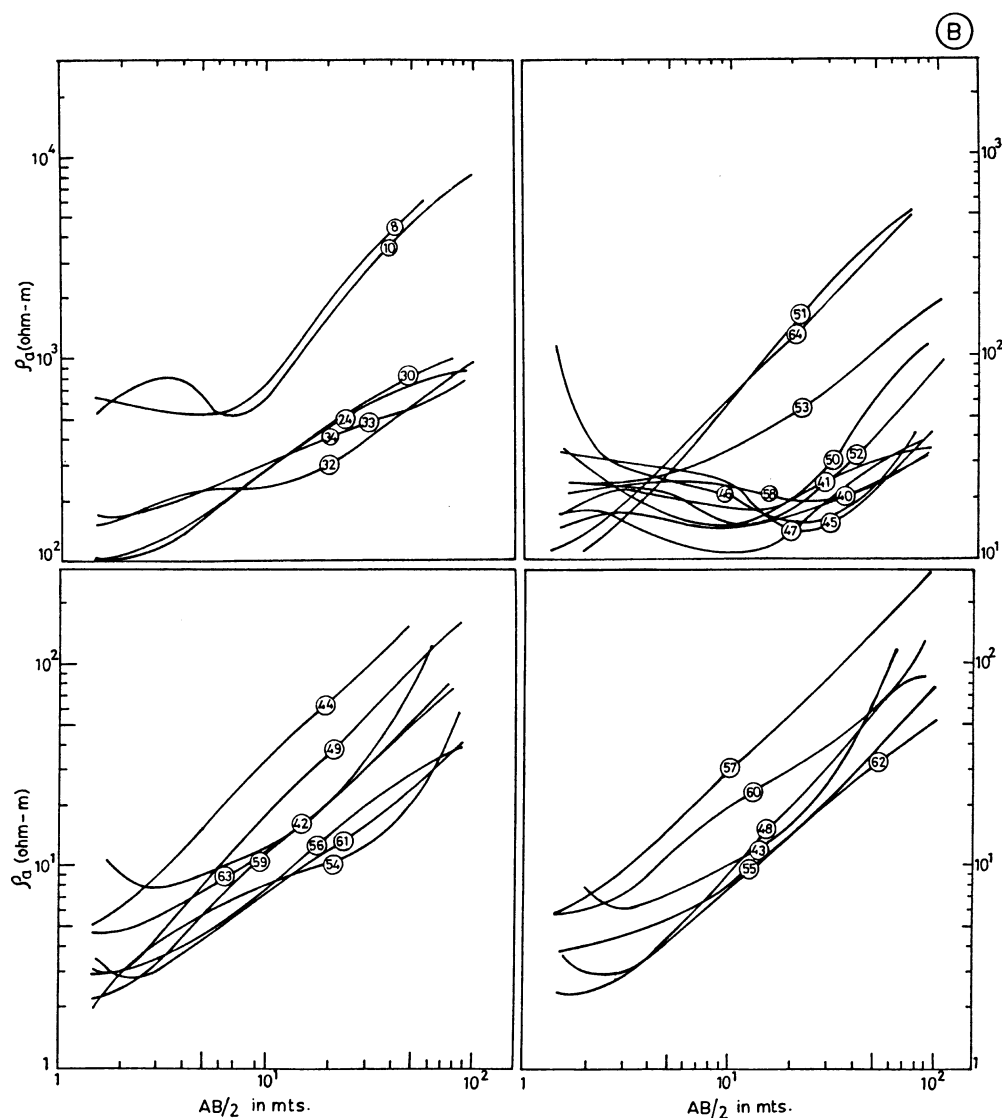


Fig. 4A-C
(Contd.)

attributed to variation in moisture content and changes in surface conditions. A three-dimensional view of apparent resistivity for the top layer is shown in Fig. 6B. There is only one peak at the northwestern part of the 3-D view that is clearly identified in the contour map, which also occurs in the quartzite boulder zone with a resistivity of 630 Ωm .

Longitudinal conductance

The ratio of different layers to their respective resistivities is known as the longitudinal conductance. The properties of a thin conducting layer can be determined in terms of longitudinal conductance, and a resistive layer can be determined by transverse resistance (Yungul 1996). Figure 7A depicts the total longitudinal conductance contour map. The conductance values in the watershed area vary from 0.01 northeast of Nereducherla village to 2.83 at Joginagaiahpalke. As the conductance increases, the resistivity naturally decreases, pointing towards groundwater potential aquifers. The 3-D view as shown in Fig. 7B depicts a few peaks, and one of them is prominently seen in the central portion.

Depth to basement

Information on the depth to bedrock for an area is a necessity in order to estimate the thickness of the zone of saturation. By using the interpreted results of the vertical electrical soundings, a depth to bedrock map has been prepared, as shown in Fig. 8. Bedrock occurs at a very shallow depth of 6.0 m towards the northern portion of the watershed, while it is deepest with a depth of 60.7 m near Kundanakota village, in the Kavulupalle plateau in the northeastern region of the watershed. The watershed area is classified into three zones based on the depth to bedrock. The deepest zone occurs in the southern and central portions of the basin, and also as isolated patches in other areas. The rest of the area is classified as a moderate zone, and the third shallow zone occurs in isolated patches, as shown in Fig. 8.

Correlation of results of soundings with lithology

The results from four resistivity probes conducted in close proximity to four wells in different formations have been correlated with the known lithologies, as shown in Fig. 9. Sounding nos. 1 and 31 were in the

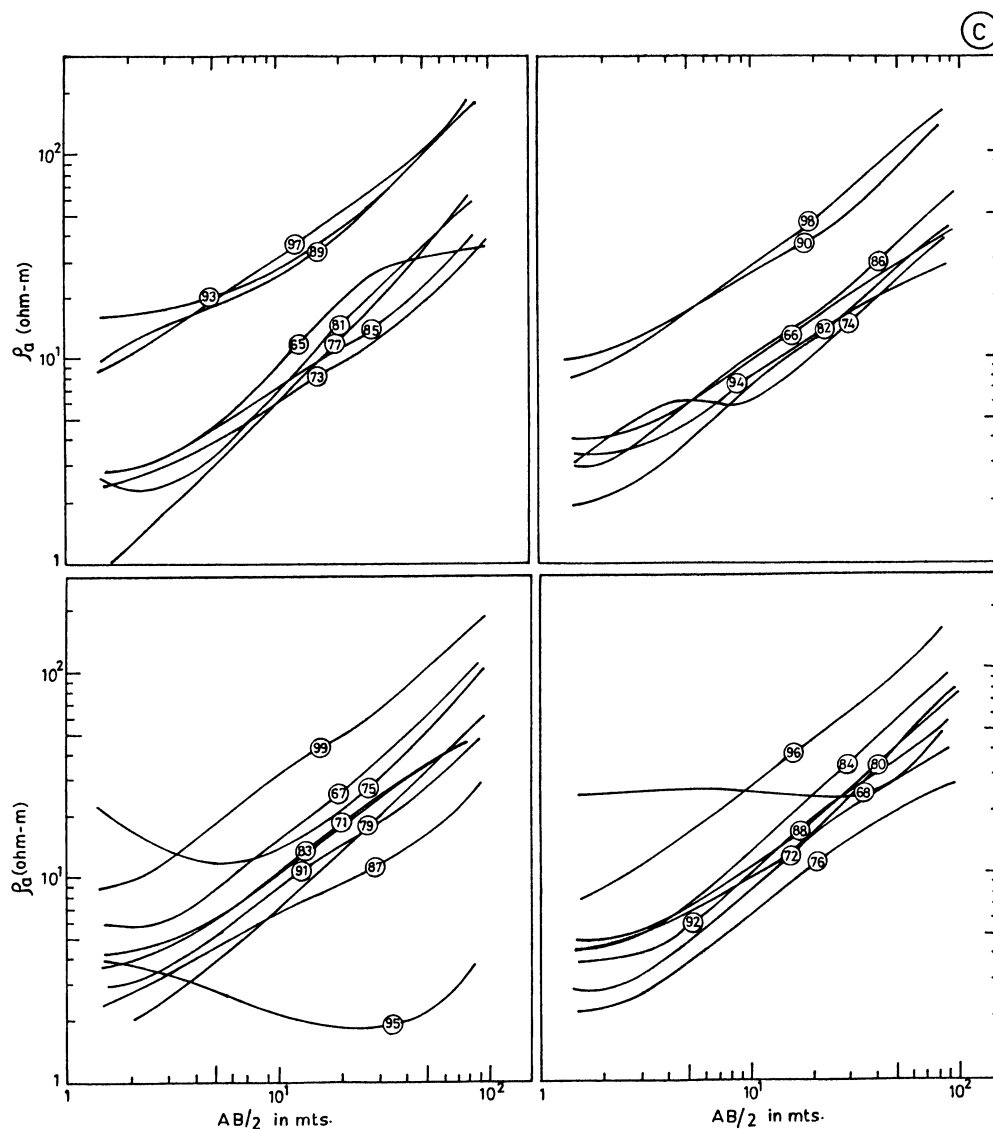


Fig. 4A-C
(Contd.)

Table 2
Analysis of vertical electrical soundings

Type of curve	Vertical electrical soundings
A-type	5,7,9,11,12,13,14,16,17,20,21,24,25,28,29,30,32,35,36,44,48,49,51,53,55,56,57,59,60,61,62,63,64,65,66,67,69,70,71,72,73,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,96,97,98,99
K-type	1,2,3,4
H-type	8,15,23,50,75
KH-type	6,10,18,26,27,37,40,41,47,54,58,68,74
HK-type	19
QH-type	45,46,52,95
HA-type	39,42,43
AK-type	22,38
AA-type	31,33,34

limestone terrain, while sounding nos. 41 and 95 were in shale.

The results from sounding no. 1 indicate four layers with absolute resistivities of 23, 897, and 228 Ωm , respectively, for the first three layers, and the bottom-most layer has very high resistivity. The depth of the

well is 55 m, and the first layer with 23 Ωm correlates to the top soil, which extends down to a depth of 1.7 m from the surface. Between 1.7–23.8 m the resistivity increases to 897 Ωm , indicating massive limestone. The third layer from 23.8–30.5 m shows a decreasing trend in resistivity, indicating fractured limestone, which

Table 3

Apparent resistivity at different depths

VES number	Longitude	Latitude	Apparent resistivity at various depths (AB/2)				Longitudinal conductance
			1.5 m	10 m	50 m	90 m	
1	77°54'12"	15°12'42"	18	120	345	308	0.10
2	77°54'50"	15°12'32"	18	120	341	295	0.13
3	77°54'33"	15°12'02"	21	161	477	390	0.07
4	77°54'20"	15°11'38"	21	160	473	377	0.07
5	77°54'46"	15°11'28"	11	48	117	214	0.40
6	77°54'38"	15°11'20"	73	68	243	459	0.15
7	77°54'32"	15°10'55"	11	49	121	226	0.30
8	77°55'32"	15°10'50"	628	755	462	9220	0.01
9	77°54'13"	15°10'30"	9	48	217	421	0.28
10	77°55'22"	15°10'16"	446	672	371	8202	0.14
11	77°55'32"	15°09'45"	35	163	946	1929	0.06
12	77°55'32"	15°09'08"	34	142	746	1666	0.07
13	77°55'38"	15°08'38"	38	166	949	1932	0.06
14	77°56'03"	15°08'17"	32	160	943	1926	0.06
15	77°54'10"	15°08'18"	26	13	25	44	0.97
16	77°55'05"	15°08'12"	8	45	228	453	0.24
17	77°55'52"	15°08'03"	10	56	254	522	0.19
18	77°53'57"	15°07'52"	35	20	21	44	1.06
19	77°53'47"	15°07'28"	33	14	28	57	1.45
20	77°55'33"	15°07'32"	9	41	134	277	0.36
21	77°56'14"	15°07'28"	9	43	141	289	0.34
22	77°53'18"	15°06'48"	2	10	36	63	1.14
23	77°54'10"	15°06'41"	33	14	25	44	1.41
24	77°55'50"	15°07'00"	107	527	686	88	0.04
25	77°56'40"	15°06'54"	107	527	692	101	0.05
26	77°57'38"	15°06'52"	100	82	200	453	0.20
27	77°58'20"	15°06'46"	100	83	188	465	0.17
28	77°58'58"	15°06'44"	48	66	217	415	0.20
29	77°53'31"	15°06'12"	2	11	70	101	1.18
30	77°55'47"	15°06'26"	101	302	726	1766	0.05
31	77°57'07"	15°06'18"	101	31	729	1640	0.73
32	77°58'06"	15°06'17"	149	259	187	779	0.07
33	77°58'53"	15°06'11"	177	334	507	710	0.06
34	77°59'36"	15°06'03"	178	335	509	723	0.07
35	77°53'12"	15°05'50"	35	58	121	264	0.41
36	77°54'07"	15°05'50"	2	11	74	113	1.05
37	77°53'10"	15°05'17"	28	57	175	314	0.25
38	77°53'58"	15°05'19"	27	56	160	295	0.22
39	77°53'36"	15°04'58"	39	20	23	38	0.75
40	77°54'36"	15°05'04"	15	11	17	31	1.11
41	77°55'18"	15°05'03"	17	12	21	44	0.90
42	77°55'57"	15°05'02"	19	12	70	38	1.05
43	77°56'40"	15°04'55"	18	9	62	25	0.72
44	78°00'04"	15°05'20"	5	40	160	270	0.39
45	77°53'48"	15°04'27"	33	20	21	38	2.52
46	77°54'26"	15°04'27"	104	20	21	38	1.87
47	77°55'09"	15°04'23"	72	10	19	44	1.87
48	77°55'54"	15°04'20"	2	8	58	94	1.20
49	77°56'40"	15°04'17"	3	15	55	207	0.64
50	77°57'27"	15°04'17"	32	14	89	107	1.05
51	78°00'35"	15°04'35"	8	62	390	559	0.19
52	77°54'13"	15°03'52"	24	13	28	107	1.17
53	77°54'53"	15°03'52"	25	18	62	220	0.38
54	77°55'40"	15°03'52"	2	6	21	57	4.29
55	77°56'33"	15°03'47"	4	8	30	88	1.69
56	77°57'32"	15°03'40"	3	6	40	38	1.71
57	78°01'15"	15°03'58"	6	40	181	271	0.41
58	77°54'56"	15°03'17"	24	25	19	38	2.65
59	77°55'40"	15°03'11"	5	11	32	57	1.13
60	77°56'20"	15°03'07"	6	18	62	145	0.78
61	77°55'28"	15°02'42"	4	7	25	44	1.18
62	77°56'25"	15°02'38"	4	8	26	50	1.18
63	77°57'17"	15°02'37"	6	14	32	101	1.12
64	78°01'20"	15°02'52"	8	62	394	572	0.38
65	77°55'42"	15°01'57"	3	7	32	88	1.57

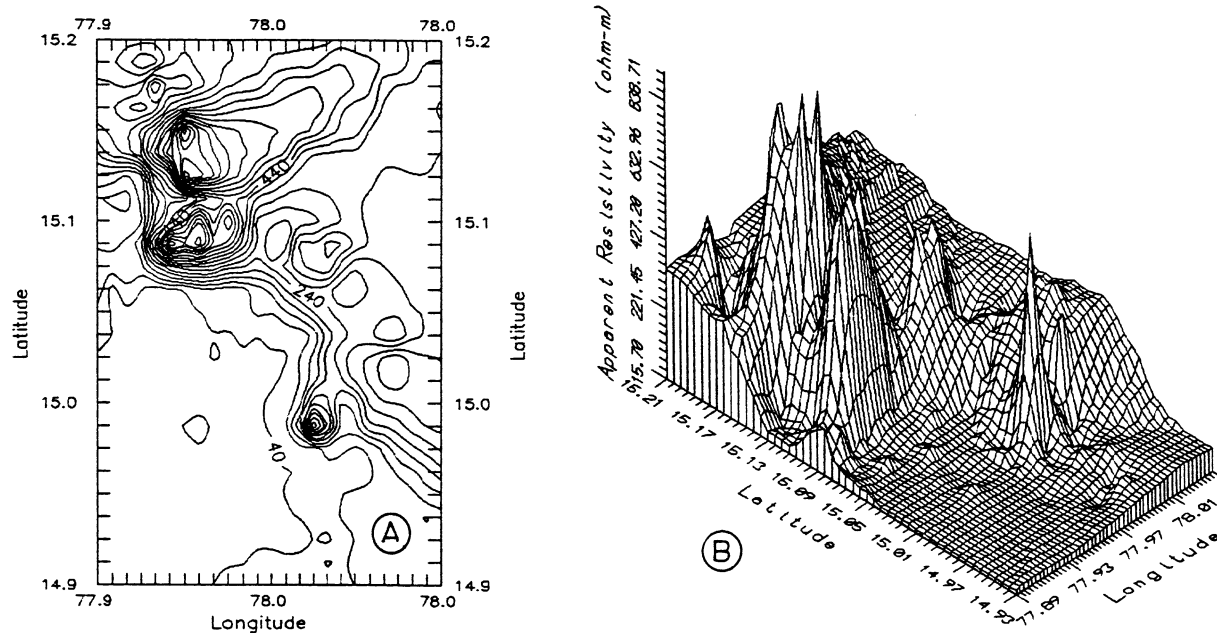
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(Contd.)

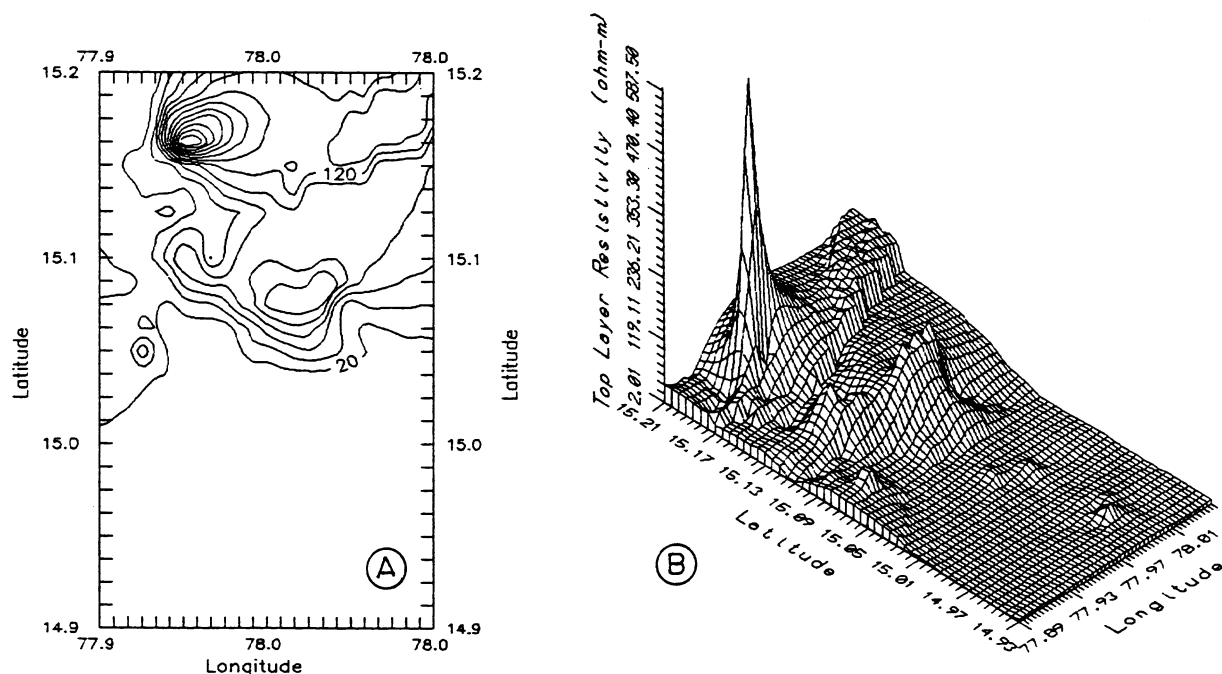
VES number	Longitude	Latitude	Apparent resistivity at various depths (AB/2)				Longitudinal conductance
			1.5 m	10 m	50 m	90 m	
66	77°56'29"	15°01'55"	4	9	13	50	1.31
67	77°57'23"	15°01'57"	6	15	36	119	0.72
68	77°58'15"	15°01'47"	25	24	25	50	1.77
69	78°01'18"	15°02'05"	9	63	395	593	0.18
70	78°02'12"	15°02'10"	9	60	337	597	0.20
71	77°56'03"	15°01'26"	4	9	17	57	1.05
72	77°56'49"	15°01'04"	3	8	19	44	1.28
73	77°57'33"	15°01'13"	2	5	17	44	3.13
74	77°58'22"	15°01'15"	3	6	28	15	1.77
75	77°59'10"	15°00'58"	21	14	679	94	0.77
76	78°00'08"	15°01'08"	2	6	24	38	1.79
77	78°02'03"	15°01'20"	9	61	341	610	0.18
78	78°02'38"	15°01'00"	10	62	342	611	0.18
79	77°56'40"	15°00'23"	3	8	32	68	1.41
80	77°58'02"	15°00'34"	3	7	32	88	1.50
81	77°59'05"	15°00'23"	3	7	28	57	1.73
82	78°00'28"	15°00'24"	2	7	23	44	1.96
83	77°57'02"	14°59'47"	4	8	28	82	0.98
84	77°58'00"	14°59'50"	5	9	32	94	0.94
85	77°58'52"	14°59'49"	3	9	28	38	2.67
86	78°00'03"	14°59'42"	3	10	36	63	1.64
87	77°57'33"	14°58'48"	4	9	17	31	2.83
88	77°58'42"	14°58'42"	5	9	55	69	1.27
89	77°59'28"	14°58'54"	10	25	92	170	0.62
90	78°00'32"	14°58'46"	10	25	87	157	0.71
91	77°58'03"	14°58'04"	3	9	32	50	2.04
92	77°59'12"	14°58'12"	5	8	51	57	1.03
93	78°00'07"	14°58'13"	15	28	104	195	0.56
94	77°58'38"	14°57'23"	2	9	17	31	1.15
95	77°59'28"	14°57'24"	40	31	21	38	2.27
96	78°00'46"	14°57'26"	8	28	92	157	0.50
97	77°59'25"	14°56'48"	9	27	91	157	0.56
98	78°00'36"	14°56'57"	8	28	94	163	0.50
99	78°01'17"	14°56'17"	9	29	95	164	0.61

Fig. 5

Apparent resistivity contour map of the area, using a Schlumberger electrode separation of 50 m (AB/2): Iso-contour map (A); 3-D representation (B)

yields significant quantities of groundwater. Similarly, the results from sounding no. 31 conducted at Kundanakota village indicate four layers, with the top soil



**Fig. 6**

Apparent resistivity contour map of the top layer: Iso-contour map (A); 3-D representation (B)

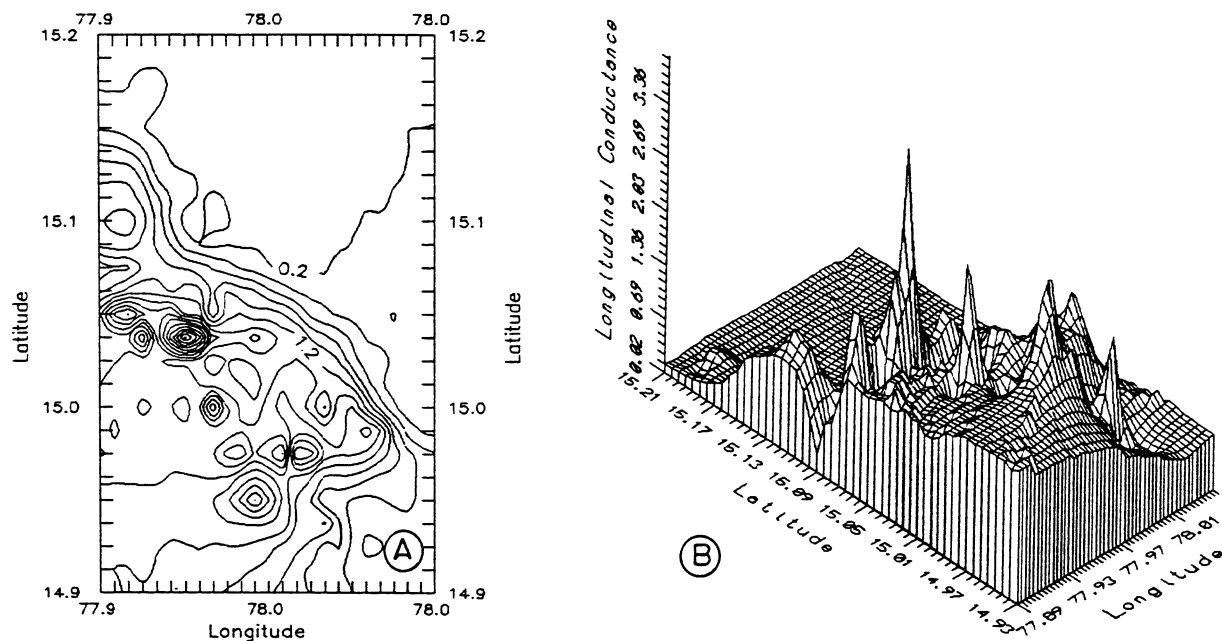
exhibiting a resistivity of $11 \Omega\text{m}$ up to a depth of 2.1 m, the second layer with a resistivity of $33 \Omega\text{m}$, indicating clay from 2.1–6.3 m, and the third layer, a fractured limestone with a resistivity of $131 \Omega\text{m}$ that occurs between 6.3–60.7 m. The well was drilled up to a depth of 65 m, yielding significant quantities of groundwater. The dense limestone at the bottom is the fourth layer with high resistivity. Limestone, though very compact, is

known to be a good water-yielding formation, due to its secondary porosity such as joints, fractures, bedding planes, and also due to being cavernous.

The results from the analysis of sounding nos. 41 and 95, at Kamalapadu and Kaverisamudram villages, respectively, suggest that their well log data corroborates perfectly with the lithology. The top soil with clay is the first layer in both bore well logs, up to a maximum of 1.7 m. Shale with red clay to a maximum of 6.3 m, exhibiting resistivities of $24 \Omega\text{m}$ and $31 \Omega\text{m}$, is the second layer. Weathered shale, highly variable in thickness from 16–38.8 m, with resistivities of $15\text{--}16 \Omega\text{m}$, is identified as the third layer. The total depth of the bore wells was 30 m and 42 m at sounding nos. 41 and 95, respectively. The bottom-most layer, with resistivities of $141 \Omega\text{m}$ and $226 \Omega\text{m}$, is hard shale.

Fig. 7

Distribution of longitudinal conductance: Iso-contour map (A); 3-D representation (B)



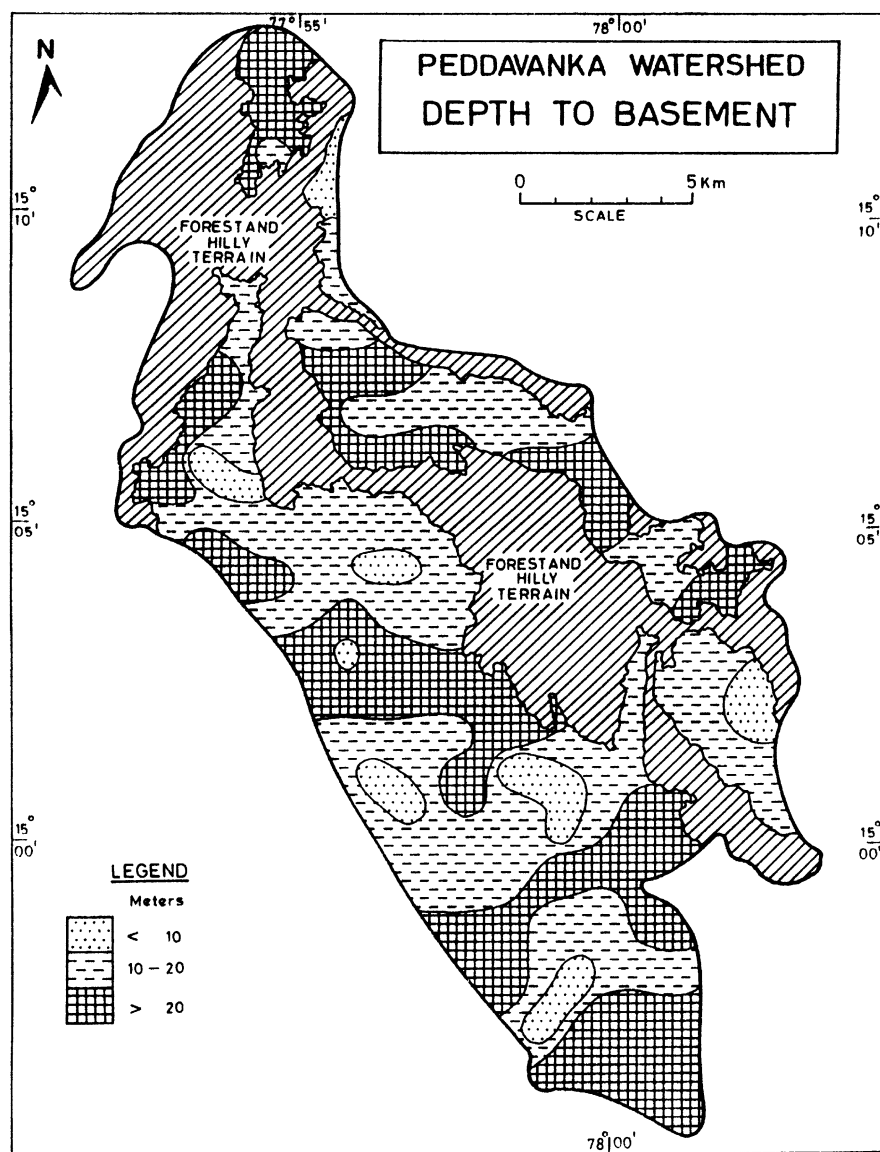


Fig. 8
The contour map showing the depth to basement

Groundwater potential aquifers

Assuming that wide variations are not present within a few kilometres, groundwater potential aquifers (Fig. 10), which are classified as good, moderate and poor zones, have been delineated by interpreting the depth to bedrock (Fig. 5), the thickness of the saturated layer (Table 1) and the resistivity of the second layer. Limestones and shales, which are devoid of primary porosity, often act as the potential aquifers when they are fractured and/or weathered (Singh and others 2003). The study reveals that the weathered and fractured portions in shale and cavernous limestone that occur in the southernmost and central portions of the watershed area contain productive water-bearing zones categorized as good potential zones. Moderate groundwater potential zones occur mainly in the western and central portions, and subordinately in the northern and southern portions of the basin, which are occupied by shale and limestone. The remaining parts of the area were demarcated as the very low saturated zone,

classified as a poor potential water zone in the dense and massive limestone and in shale.

Conclusions

Through investigations conducted by the electrical resistivity survey, groundwater potential aquifers have been delineated. Fractured and weathered horizons have been identified in the study area underlying VES stations, and all of these constitute good aquifer zones. Good prospects therefore exist for groundwater development in the study area where the depth to basement is relatively thick and has favourable low resistivity values, while those with thin depth to bedrock and high resistivity value have a lower potential for an aquifer. The interpretation of depth to bedrock is in good

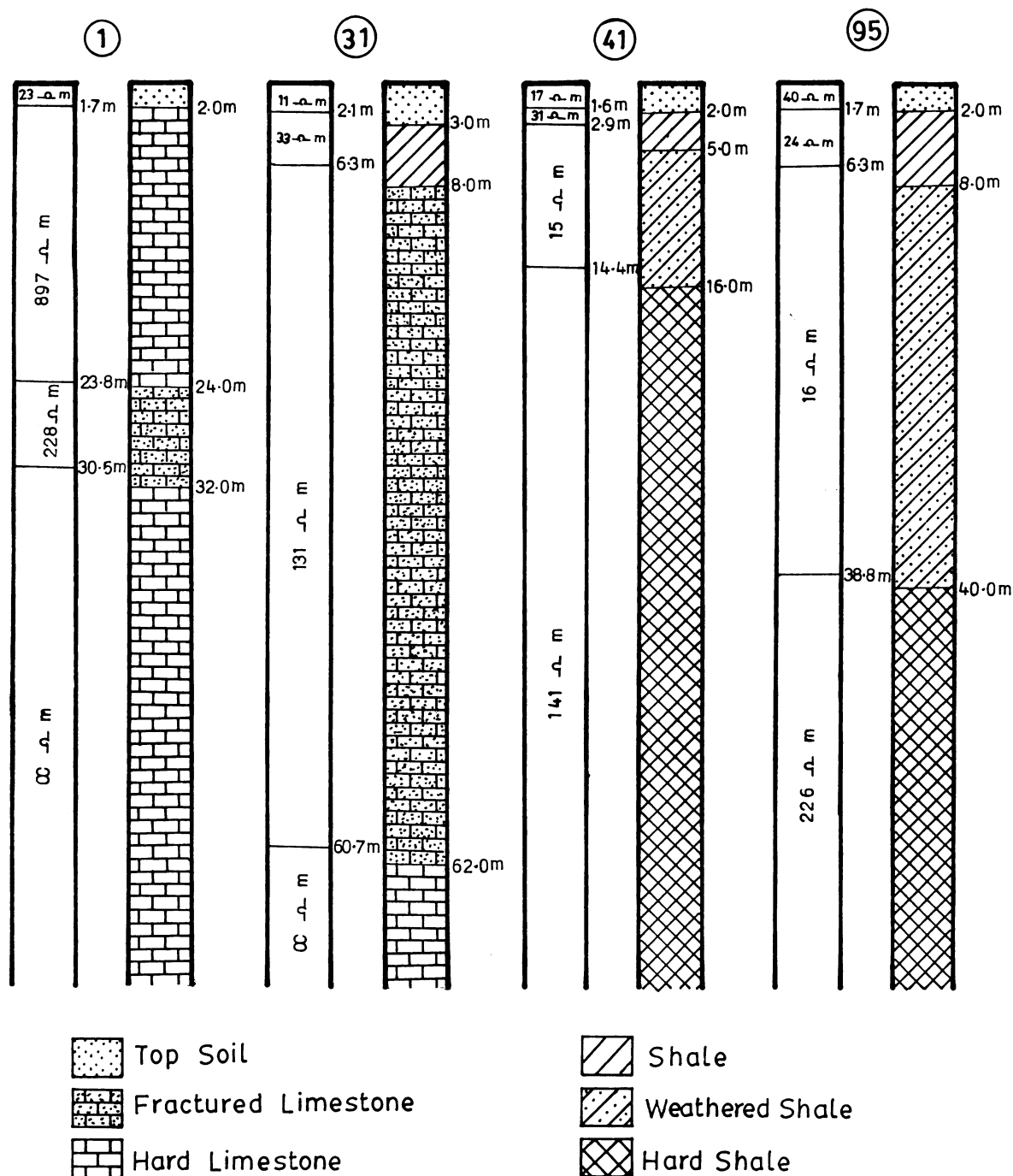


Fig. 9

Correlation of resistivity results with bore hole data

agreement with the actual depth observed in the dug wells, dug-cum-bore wells and bore wells. The limestone, owing to its cavernous nature, produces high yields that occur as springs at a few locations in the watershed on the one hand, while it is very hard and massive without any weathered portion and is totally devoid of water at some places on the other hand.

The electrical resistivity data therefore gives reasonably accurate results that can be used to understand the

subsurface layers and basement configuration in ground-water exploration.

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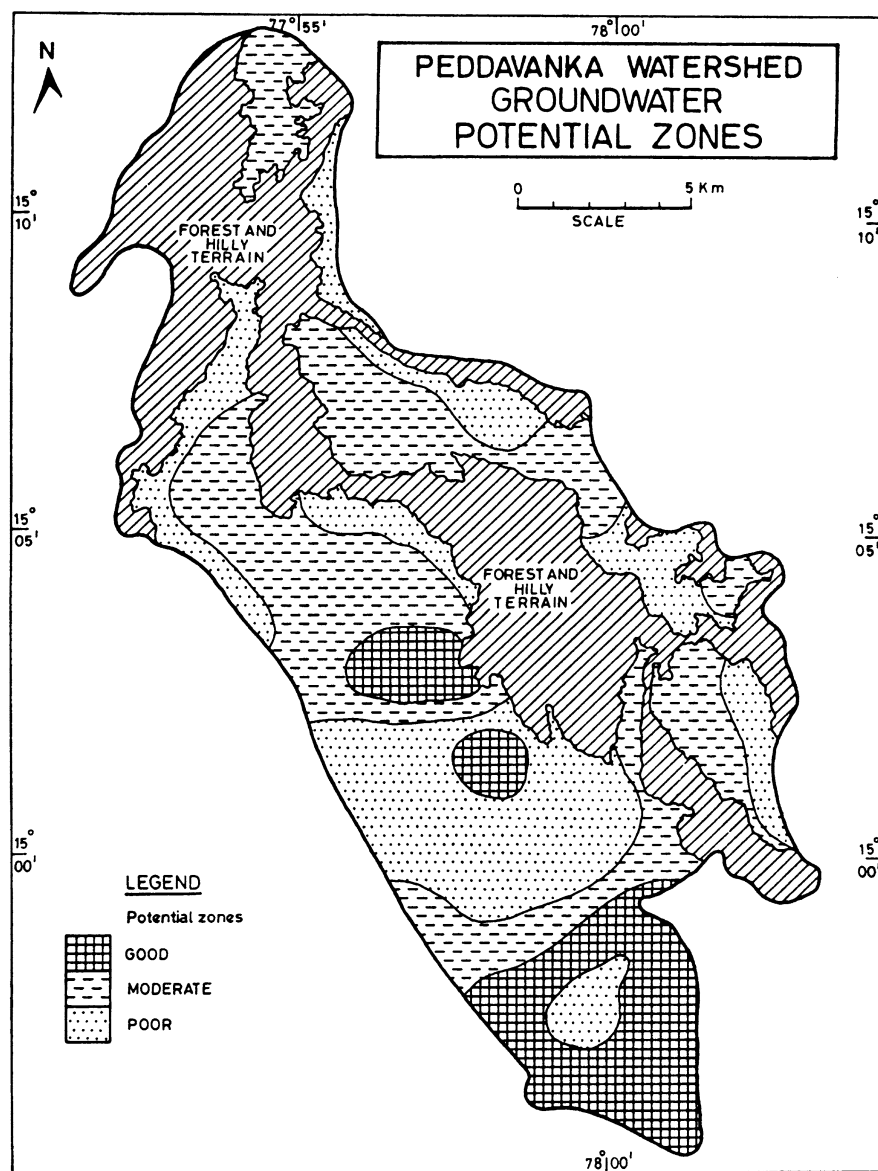


Fig. 10
Map showing groundwater potential zones

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